Smart Health and Fitness Monitor App

Mini Project Report

Submitted by

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Submitted to

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Development Environment: Android Studio

Programming Language: Java and XML

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1. Introduction and Objective

Smart Health and Fitness Monitor is a native Android application developed to encourage a healthy and active lifestyle. The application collects basic profile information, estimates physical movement using the phone accelerometer, stores activity history locally, lets the user set a personal daily step goal, and provides a notification when that goal is reached.

The application is designed as a simple student mini project rather than a medical device. Step totals are estimates derived from sensor readings and should not be treated as medical measurements.

Objectives

- Create and validate a user profile containing name, age, gender, height, and weight.
- Estimate steps from accelerometer readings while activity monitoring is active.
- Save daily activity information, including date, time, estimated steps, and duration, in an SQLite database.
- Allow the user to set and retain a daily step goal using SharedPreferences.
- Show a motivational Android notification once the configured goal is achieved.

2. Application Design

The application uses three focused screens so that the profile, dashboard, and history are not crowded onto one page. A dark brick background is used with light rounded cards and teal headings to create a clear fitness-themed visual style.

User Interface Design

Screen	Purpose
Profile screen	Collects the user name, age, gender, height, and weight. Validation prevents an incomplete profile from continuing.
Dashboard	Displays the daily goal, current estimated steps, duration, monitoring status, and actions to change the goal, save an activity, and open prior records.
Records screen	Lists saved records in newest-first order with date, time, steps, and activity duration.

Android Components Used

Component	Use in the app
Activity	MainActivity, DashboardActivity, and RecordsActivity manage the three application screens.
Views and XML layouts	EditText, RadioGroup, RadioButton, TextView, Button, ScrollView, and LinearLayout build the interface.
SharedPreferences	Stores profile values and the chosen daily step goal after the application restarts.
SensorManager and accelerometer	Receives movement readings and applies a simple magnitude threshold to estimate steps.
SQLiteOpenHelper	Creates and accesses the local SQLite activity-record database.
NotificationChannel and NotificationCompat	Creates and displays the daily-goal achievement notification.

3. Implementation Explanation

3.1 User Profile Management

The profile screen uses EditText controls for name, age, height, and weight, and a RadioGroup for gender. When Save and Continue is pressed, the app checks that every field has a value and that age is positive. Valid data is stored in SharedPreferences and then the dashboard is opened.

3.2 Physical Activity Monitoring

DashboardActivity registers an accelerometer listener after the user presses Start Activity Monitoring. The app calculates the magnitude of the x, y, and z sensor values. A magnitude crossing the selected threshold with a short time gap is counted as an estimated step. The listener is unregistered in onPause and registered again in onResume when monitoring is active.

3.3 Daily Activity Records

DatabaseHelper extends SQLiteOpenHelper and creates an activity_records table. Each saved record contains the date and time, estimated steps, and activity duration in minutes. RecordsActivity retrieves the records in descending identifier order so the newest record appears first.

3.4 Daily Goal Settings

The dashboard displays the goal saved in SharedPreferences. The Change Daily Goal button opens an AlertDialog containing a numeric input field. The new goal is validated, saved, and immediately displayed on the dashboard.

3.5 Activity Notifications

When the estimated step count reaches the saved goal, the app creates one high-priority notification for the monitoring session. Android 13 and later devices are asked for POST_NOTIFICATIONS permission before notifications can be shown.

4. Application Screenshots

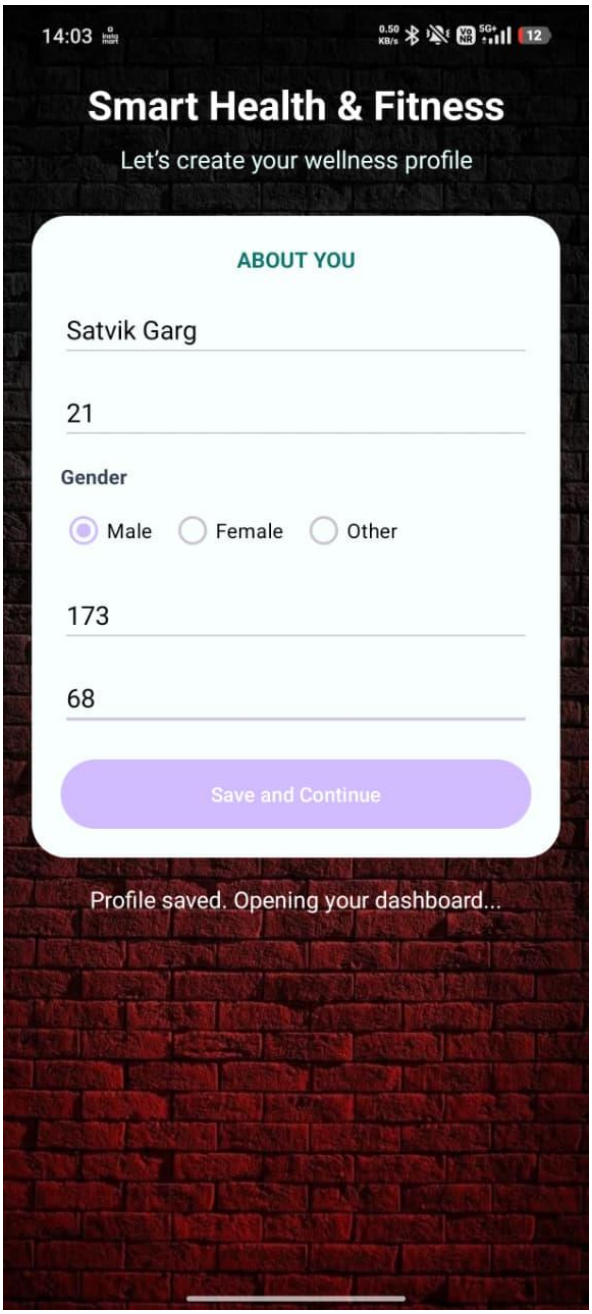


Figure 1. Profile screen with saved user details and validation-ready controls.

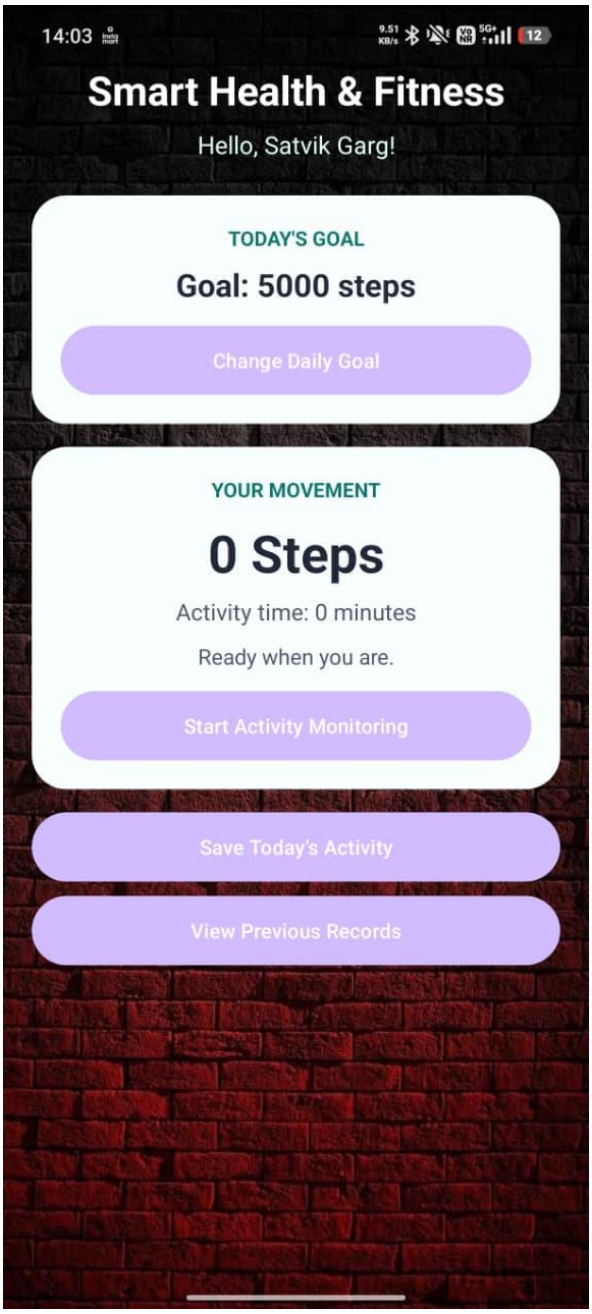


Figure 2. Dashboard showing the configured 5000-step goal and activity controls.

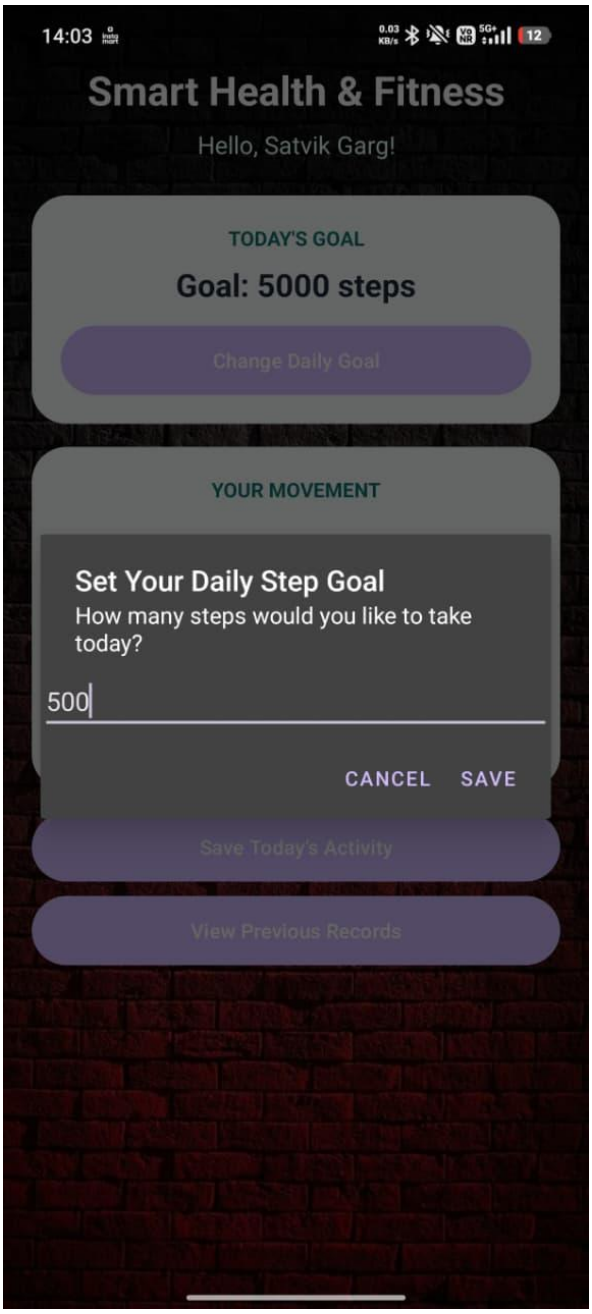


Figure 3. Daily step-goal input dialog.

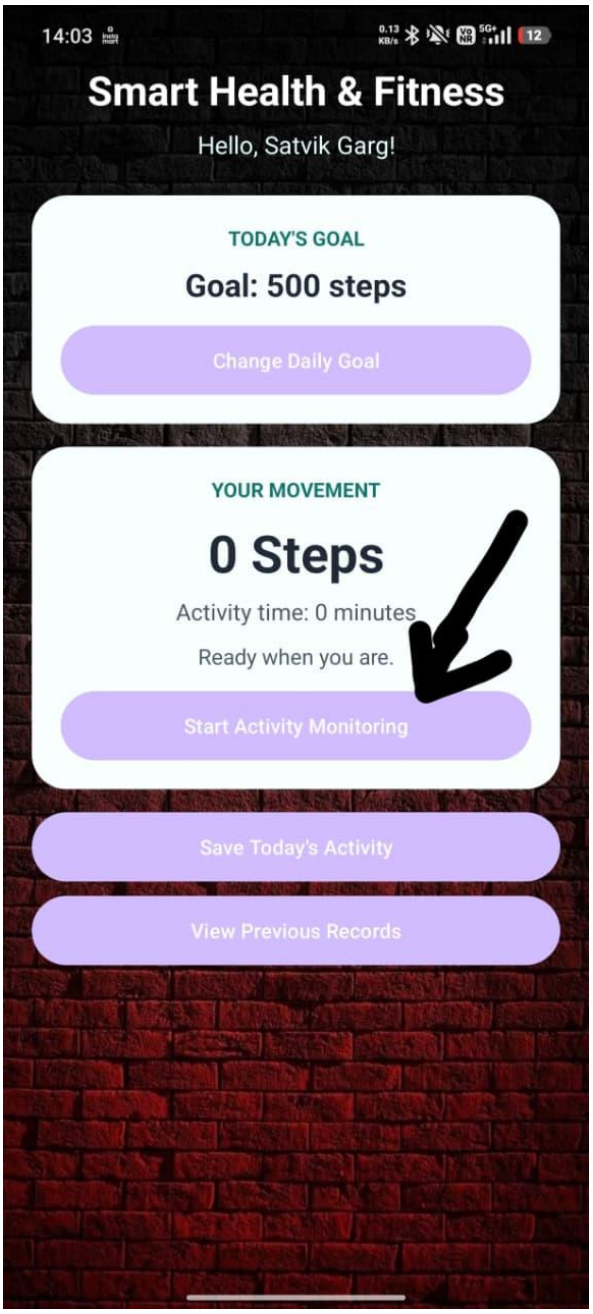


Figure 4. Dashboard action for starting activity monitoring.

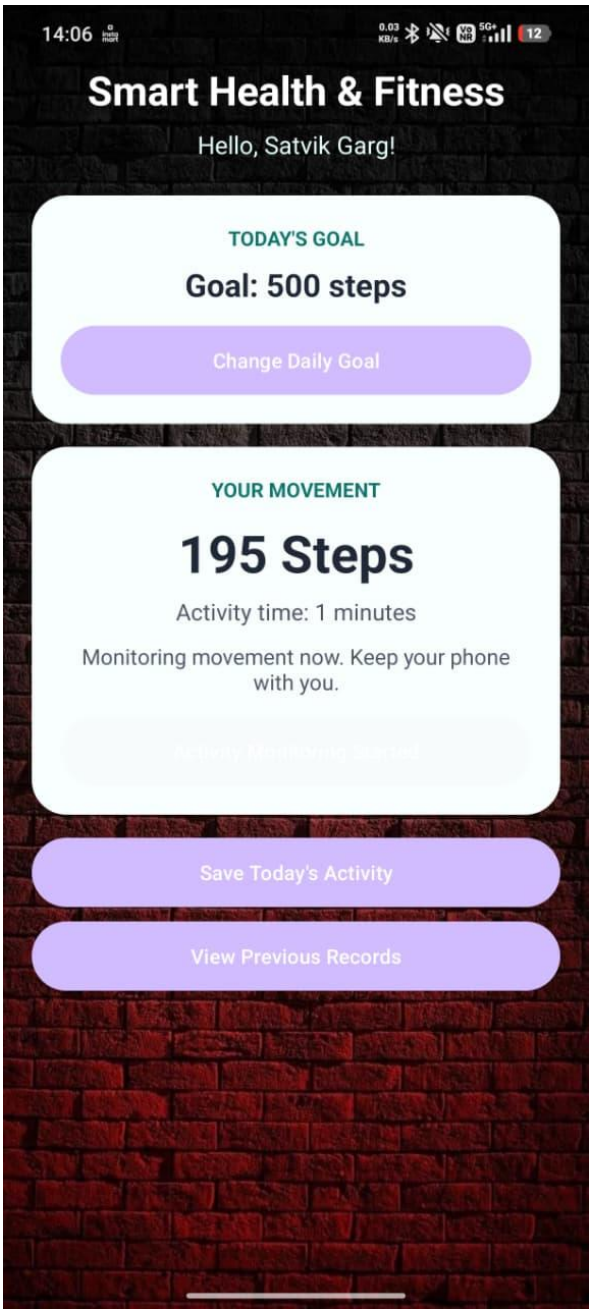


Figure 5. Active monitoring with 195 estimated steps and one minute of activity time.

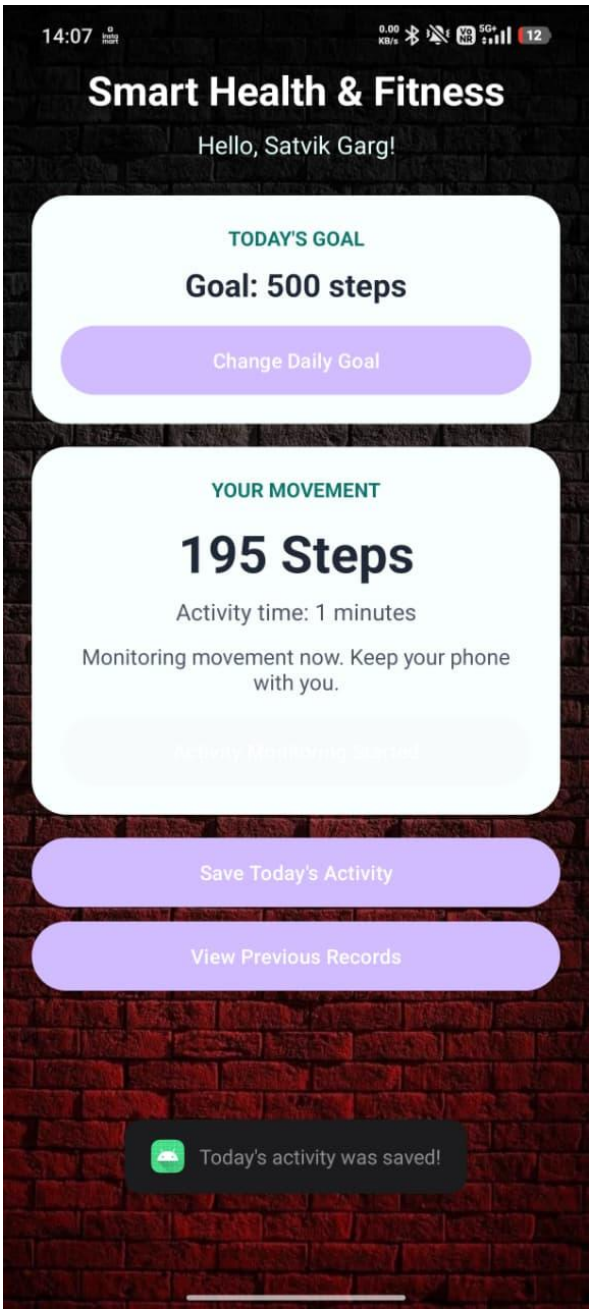


Figure 6. Confirmation that the current activity record was saved.

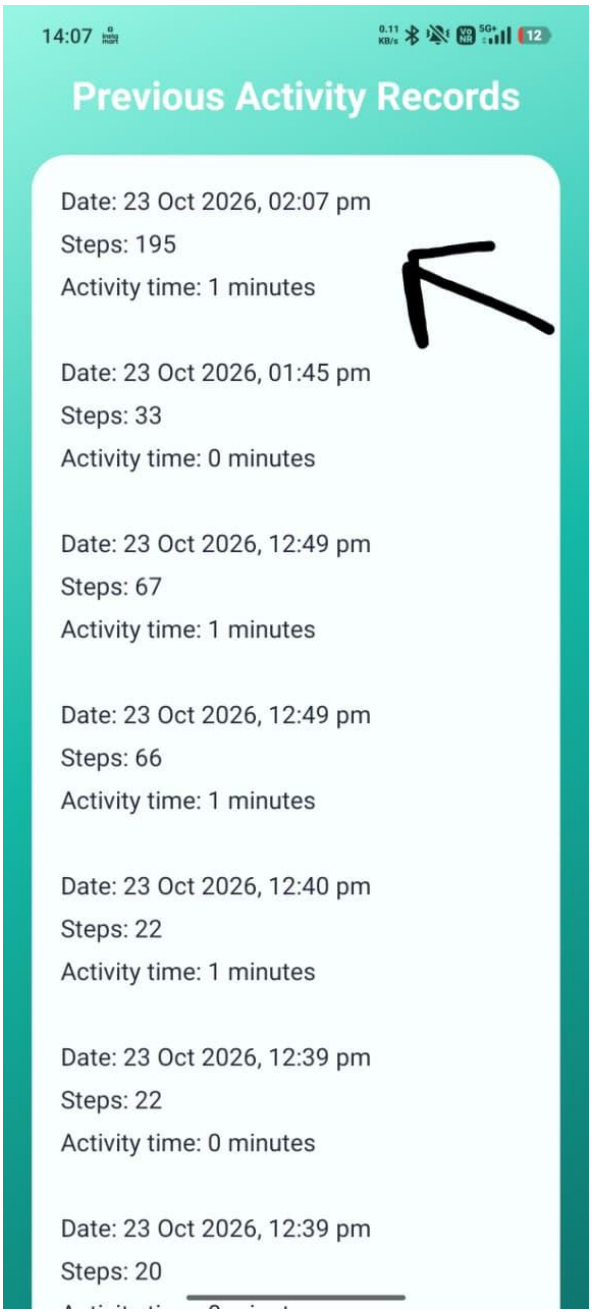


Figure 7. Previous activity records with date, time, steps, and duration.

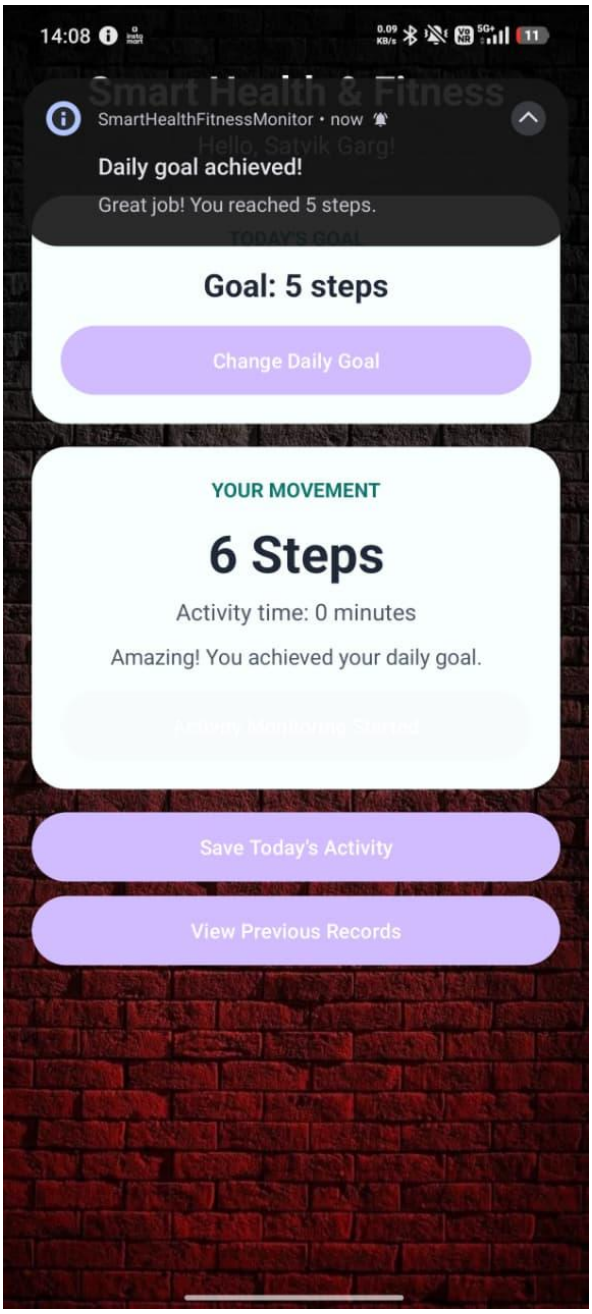


Figure 8. Daily-goal notification after the 5-step goal is achieved.

5. Testing and Results

The following functional tests were performed on the application. The captured screenshots demonstrate the successful results for the main user flows.

ID	Feature	Test action	Expected result	Actual result
T1	Save profile	Enter valid profile values and tap Save and Continue.	Profile is saved and dashboard opens.	Pass. Profile saved message appeared and dashboard opened.
T2	Set goal	Open Change Daily Goal, enter 500, and save.	Dashboard displays Goal: 500 steps.	Pass. The dashboard updated to 500 steps.
T3	Monitor activity	Start activity monitoring and move with the phone.	Estimated step count and duration update.	Pass. Screenshot shows 195 steps and one minute.
T4	Save record	Tap Save Today's Activity after monitoring.	A local activity record is saved.	Pass. A save confirmation message appeared.
T5	View records	Open View Previous Records.	Stored records show date, time, steps, and duration.	Pass. Previous records were displayed.
T6	Goal notification	Set the goal to 5 and continue monitoring until the count reaches it.	A daily-goal notification appears.	Pass. Notification shown after 5 steps were reached.

6. Complete Source Code

This section contains the Java and XML source files used to implement the application. The image asset fitness_background.jpg is included in the Android drawable resources folder and is referenced by the layout files.

MainActivity.java

```
package com.example.smarthealthfitnessmonitor;

import android.content.Intent;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText editName, editAge, editHeight, editWeight;
    RadioGroup radioGender;
    Button btnSaveProfile;
    TextView textWelcome;
    SharedPreferences preferences;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        editName = findViewById(R.id.editName);
        editAge = findViewById(R.id.editAge);
        editHeight = findViewById(R.id.editHeight);
        editWeight = findViewById(R.id.editWeight);
        radioGender = findViewById(R.id.radioGender);
        btnSaveProfile = findViewById(R.id.btnSaveProfile);
        textWelcome = findViewById(R.id.textWelcome);

        preferences = getSharedPreferences("UserProfile", MODE_PRIVATE);

        loadSavedProfile();

        btnSaveProfile.setOnClickListener(view -> saveProfileAndOpenDashboard());
    }

    private void saveProfileAndOpenDashboard() {
        String name = editName.getText().toString().trim();
        String age = editAge.getText().toString().trim();
        String height = editHeight.getText().toString().trim();
        String weight = editWeight.getText().toString().trim();

        int selectedGenderId = radioGender.getCheckedRadioButtonId();

        if (name.isEmpty() || age.isEmpty() || height.isEmpty() || weight.isEmpty()
                || selectedGenderId == -1) {
            Toast.makeText(this,
                    "Please fill in every box and choose a gender.",
                    Toast.LENGTH_SHORT).show();
            return;
        }

        int ageNumber = Integer.parseInt(age);

        if (ageNumber <= 0) {
            editAge.setError("Age must be bigger than 0.");
```

```
            return;
        }

        RadioButton selectedGender = findViewById(selectedGenderId);
        String gender = selectedGender.getText().toString();

        SharedPreferences.Editor editor = preferences.edit();
        editor.putString("name", name);
        editor.putString("age", age);
        editor.putString("gender", gender);
        editor.putString("height", height);
        editor.putString("weight", weight);
        editor.apply();

        textWelcome.setText("Profile saved. Opening your dashboard...");
        Toast.makeText(this, "Profile saved!", Toast.LENGTH_SHORT).show();

        Intent intent = new Intent(MainActivity.this, DashboardActivity.class);
        startActivity(intent);
    }

    private void loadSavedProfile() {
        String savedName = preferences.getString("name", "");

        if (!savedName.isEmpty()) {
            editName.setText(savedName);
            editAge.setText(preferences.getString("age", ""));
            editHeight.setText(preferences.getString("height", ""));

            editWeight.setText(preferences.getString("weight", ""));

            String savedGender = preferences.getString("gender", "");

            if (savedGender.equals("Male")) {
                radioGender.check(R.id.radioMale);
            } else if (savedGender.equals("Female")) {
                radioGender.check(R.id.radioFemale);
            } else if (savedGender.equals("Other")) {
                radioGender.check(R.id.radioOther);
            }

            textWelcome.setText("Welcome back, " + savedName + "!");
        }
    }
}
```

DashboardActivity.java

```
package com.example.smarthealthfitnessmonitor;

import android.app.AlertDialog;
import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.pm.PackageManager;
import android.os.Build;
import android.content.SharedPreferences;
import android.content.Intent;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.text.InputType;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.content.ContextCompat;

import java.text.SimpleDateFormat;
import java.util.Date;
import java.util.Locale;

public class DashboardActivity extends AppCompatActivity implements SensorEventListener {
    private static final String CHANNEL_ID = "fitness_goal_channel";
    private boolean goalNotificationShown = false;
    TextView textDashboardGreeting, textDashboardGoal, textStepCount;
    TextView textDuration, textSensorStatus;
    Button btnEditGoal, btnStartMonitoring, btnSaveActivity, btnViewRecords;

    SharedPreferences preferences;
    DatabaseHelper databaseHelper;

    SensorManager sensorManager;
    Sensor accelerometer;

    boolean monitoring = false;
    int stepCount = 0;
    float previousMagnitude = 0;
    long lastStepTime = 0;
    long monitoringStartTime = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_dashboard);

        textDashboardGreeting = findViewById(R.id.textDashboardGreeting);
        textDashboardGoal = findViewById(R.id.textDashboardGoal);
        textStepCount = findViewById(R.id.textStepCount);
        textDuration = findViewById(R.id.textDuration);
        textSensorStatus = findViewById(R.id.textSensorStatus);

        btnEditGoal = findViewById(R.id.btnEditGoal);
        btnStartMonitoring = findViewById(R.id.btnStartMonitoring);
        btnSaveActivity = findViewById(R.id.btnSaveActivity);
        btnViewRecords = findViewById(R.id.btnViewRecords);

        preferences = getSharedPreferences("UserProfile", MODE_PRIVATE);
        createNotificationChannel();
        requestNotificationPermission();
        databaseHelper = new DatabaseHelper(this);

        loadUserDetails();

        sensorManager = (SensorManager) getSystemService(SENSOR_SERVICE);
```

```
    accelerometer = sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER);

    if (accelerometer == null) {
        textSensorStatus.setText("This device does not have a movement sensor.");
        btnStartMonitoring.setEnabled(false);
    }

    btnEditGoal.setOnClickListener(view -> showGoalBox());
    btnStartMonitoring.setOnClickListener(view -> startMonitoring());
    btnSaveActivity.setOnClickListener(view -> saveTodayActivity());

    btnViewRecords.setOnClickListener(view -> {
        Intent intent = new Intent(DashboardActivity.this, RecordsActivity.class);
        startActivity(intent);
    });
}

private void loadUserDetails() {
    String name = preferences.getString("name", "Friend");
    int goal = preferences.getInt("stepGoal", 0);

    textDashboardGreeting.setText("Hello, " + name + "!");
    textDashboardGoal.setText("Goal: " + goal + " steps");
}

private void showGoalBox() {
    EditText goalInput = new EditText(this);
    goalInput.setInputType(InputType.TYPE_CLASS_NUMBER);
    goalInput.setHint("For example: 5000");

    new AlertDialog.Builder(this)
            .setTitle("Set Your Daily Step Goal")
            .setMessage("How many steps would you like to take today?")
            .setView(goalInput)
            .setPositiveButton("Save", (dialog, which) -> {
                String goalText = goalInput.getText().toString().trim();

                if (goalText.isEmpty()) {
                    Toast.makeText(this, "Please enter a number.", Toast.LENGTH_SHORT).show();
                    return;
                }

                int goal = Integer.parseInt(goalText);

                if (goal <= 0) {
                    Toast.makeText(this, "Goal must be bigger than 0.", Toast.LENGTH_SHORT).show();
                    return;
                }

                preferences.edit().putInt("stepGoal", goal).apply();
                textDashboardGoal.setText("Goal: " + goal + " steps");
                Toast.makeText(this, "Daily goal saved!", Toast.LENGTH_SHORT).show();
            })
            .setNegativeButton("Cancel", null)
            .show();
}

private void startMonitoring() {
    if (accelerometer == null) {
        return;
    }

    monitoring = true;
    monitoringStartTime = System.currentTimeMillis();

    sensorManager.registerListener(
            this,
            accelerometer,
            SensorManager.SENSOR_DELAY_NORMAL
    );

    textSensorStatus.setText("Monitoring movement now. Keep your phone with you.");
    btnStartMonitoring.setText("Activity Monitoring Started");
    btnStartMonitoring.setEnabled(false);
}
```

```
private void saveTodayActivity() {
    long durationMilliseconds = 0;

    if (monitoringStartTime > 0) {
        durationMilliseconds = System.currentTimeMillis() - monitoringStartTime;
    }

    int durationMinutes = (int) (durationMilliseconds / 60000);

    String todayDate = new SimpleDateFormat(
            "dd MMM yyyy, hh:mm a",
            Locale.getDefault()
    ).format(new Date());

    databaseHelper.saveActivity(todayDate, stepCount, durationMinutes);

    Toast.makeText(
            this,
            "Today's activity was saved!",
            Toast.LENGTH_SHORT
    ).show();
}

@Override
public void onSensorChanged(SensorEvent event) {
    if (!monitoring) {
        return;
    }

    float x = event.values[0];
    float y = event.values[1];
    float z = event.values[2];

    float magnitude = (float) Math.sqrt(x * x + y * y + z * z);
    long currentTime = System.currentTimeMillis();

    if (magnitude > 12 && previousMagnitude <= 12
            && currentTime - lastStepTime > 350) {

        stepCount++;
        lastStepTime = currentTime;
        textStepCount.setText(stepCount + " Steps");
        checkGoalReached();
    }

    previousMagnitude = magnitude;
    updateDuration();
}

private void updateDuration() {
    long milliseconds = System.currentTimeMillis() - monitoringStartTime;
    long minutes = milliseconds / 60000;
    textDuration.setText("Activity time: " + minutes + " minutes");
}

@Override
public void onAccuracyChanged(Sensor sensor, int accuracy) {
    // Nothing is needed here.
}

@Override
protected void onPause() {
    super.onPause();

    if (monitoring) {
        sensorManager.unregisterListener(this);
    }
}

@Override
protected void onResume() {
    super.onResume();
```

```
        if (monitoring && accelerometer != null) {
            sensorManager.registerListener(
                    this,
                    accelerometer,
                    SensorManager.SENSOR_DELAY_NORMAL
            );
        }
    }
    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            NotificationChannel channel = new NotificationChannel(
                    CHANNEL_ID,
                    "Fitness Goal Notifications",
                    NotificationManager.IMPORTANCE_HIGH
            );

            NotificationManager manager = getSystemService(NotificationManager.class);
            manager.createNotificationChannel(channel);
        }
    }

    private void requestNotificationPermission() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (ContextCompat.checkSelfPermission(
                    this,
                    Manifest.permission.POST_NOTIFICATIONS
            ) != PackageManager.PERMISSION_GRANTED) {

                ActivityCompat.requestPermissions(
                        this,
                        new String[]{Manifest.permission.POST_NOTIFICATIONS},
                        100
                );
            }
        }
    }

    private void checkGoalReached() {
        int goal = preferences.getInt("stepGoal", 0);

        if (goal > 0 && stepCount >= goal && !goalNotificationShown) {
            goalNotificationShown = true;

            NotificationCompat.Builder builder =
                    new NotificationCompat.Builder(this, CHANNEL_ID)
                            .setSmallIcon(android.R.drawable.ic_dialog_info)
                            .setContentTitle("Daily goal achieved!")
                            .setContentText("Great job! You reached " + goal + " steps.")
                            .setPriority(NotificationCompat.PRIORITY_HIGH)
                            .setAutoCancel(true);

            NotificationManager manager =
                    (NotificationManager) getSystemService(NOTIFICATION_SERVICE);

            manager.notify(1, builder.build());

            textSensorStatus.setText("Amazing! You achieved your daily goal.");
        }
    }
}
```

RecordsActivity.java

```
package com.example.smarthealthfitnessmonitor;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class RecordsActivity extends AppCompatActivity {

    TextView textRecords;
    Button btnBackDashboard;
    DatabaseHelper databaseHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_records);

        textRecords = findViewById(R.id.textRecords);
        btnBackDashboard = findViewById(R.id.btnBackDashboard);

        databaseHelper = new DatabaseHelper(this);

        String records = databaseHelper.getAllActivityRecords();
        textRecords.setText(records);

        btnBackDashboard.setOnClickListener(view -> finish());
    }
}
```

DatabaseHelper.java

```
package com.example.smarthealthfitnessmonitor;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "FitnessDatabase.db";
    private static final int DATABASE_VERSION = 1;

    public static final String TABLE_ACTIVITY = "activity_records";
    public static final String COLUMN_ID = "id";
    public static final String COLUMN_DATE = "record_date";
    public static final String COLUMN_STEPS = "steps";
    public static final String COLUMN_DURATION = "duration";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, DATABASE_VERSION);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String createTable = "CREATE TABLE " + TABLE_ACTIVITY + " ("
                + COLUMN_ID + " INTEGER PRIMARY KEY AUTOINCREMENT, "
                + COLUMN_DATE + " TEXT, "
                + COLUMN_STEPS + " INTEGER, "
                + COLUMN_DURATION + " INTEGER)";

        db.execSQL(createTable);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_ACTIVITY);
        onCreate(db);
    }

    public void saveActivity(String date, int steps, int duration) {
        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues values = new ContentValues();
        values.put(COLUMN_DATE, date);
        values.put(COLUMN_STEPS, steps);
        values.put(COLUMN_DURATION, duration);

        db.insert(TABLE_ACTIVITY, null, values);
        db.close();
    }

    public String getAllActivityRecords() {
        SQLiteDatabase db = this.getReadableDatabase();

        Cursor cursor = db.query(
                TABLE_ACTIVITY,
                null,
                null,
                null,
                null,
                null,
                COLUMN_ID + " DESC"
        );

        StringBuilder records = new StringBuilder();

        if (cursor.getCount() == 0) {
            records.append("No activity records have been saved yet.");
        } else {
            while (cursor.moveToNext()) {
                String date = cursor.getString(
```

```
                        cursor.getColumnIndexOrThrow(COLUMN_DATE)
                );

                int steps = cursor.getInt(
                        cursor.getColumnIndexOrThrow(COLUMN_STEPS)
                );

                int duration = cursor.getInt(
                        cursor.getColumnIndexOrThrow(COLUMN_DURATION)
                );

                records.append("Date: ").append(date).append("\n");
                records.append("Steps: ").append(steps).append("\n");
                records.append("Activity time: ")
                        .append(duration)
                        .append(" minutes\n\n");
            }
        }

        cursor.close();
        db.close();

        return records.toString();
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">
    <uses-permission android:name="android.permission.POST_NOTIFICATIONS" />

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SmartHealthFitnessMonitor">
        <activity
            android:name=".MainActivity"
            android:exported="true"
            android:windowSoftInputMode="adjustResize">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity
            android:name=".DashboardActivity"
            android:exported="false"
            android:windowSoftInputMode="adjustResize" />
        <activity
            android:name=".RecordsActivity"
            android:exported="false"
            android:windowSoftInputMode="adjustResize" />
    </application>

</manifest>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/fitness_background"
    android:fillViewport="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Smart Health &amp;amp; Fitness"
            android:textColor="#FFFFFF"
            android:textSize="28sp"
            android:textStyle="bold"
            android:gravity="center"
            android:layout_marginTop="30dp" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Let’s create your wellness profile"
            android:textColor="#E6FFFB"
            android:textSize="17sp"
            android:gravity="center"
            android:layout_marginTop="8dp"
            android:layout_marginBottom="28dp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/bg_white_card">

            <TextView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="ABOUT YOU"
                android:textColor="#0F766E"
                android:textSize="15sp"
                android:textStyle="bold"
                android:gravity="center"
                android:layout_marginBottom="16dp" />

            <EditText
                android:id="@+id/editName"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Your name"
                android:textColor="#000000"
                android:inputType="textPersonName"
                android:layout_marginBottom="12dp" />

            <EditText
                android:id="@+id/editAge"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Your age"
                android:textColor="#000000"
                android:inputType="number"
                android:layout_marginBottom="12dp" />

            <TextView
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Gender"
                android:textColor="#374151"
                android:textStyle="bold"
                android:layout_marginBottom="4dp" />
```

```
            <RadioGroup
                android:id="@+id/radioGender"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:orientation="horizontal"
                android:layout_marginBottom="12dp">

                <RadioButton
                    android:id="@+id/radioMale"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Male"
                    android:textColor="#000000"
                    />

                <RadioButton
                    android:id="@+id/radioFemale"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Female"
                    android:textColor="#000000"
                    android:layout_marginStart="12dp" />

                <RadioButton
                    android:id="@+id/radioOther"
                    android:layout_width="wrap_content"
                    android:layout_height="wrap_content"
                    android:text="Other"
                    android:textColor="#000000"
                    android:layout_marginStart="12dp" />

            </RadioGroup>

            <EditText
                android:id="@+id/editHeight"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Height in cm"
                android:textColor="#000000"
                android:inputType="numberDecimal"
                android:layout_marginBottom="12dp" />

            <EditText
                android:id="@+id/editWeight"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:hint="Weight in kg"
                android:textColor="#000000"
                android:inputType="numberDecimal"
                android:layout_marginBottom="20dp" />

            <Button
                android:id="@+id/btnSaveProfile"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Save and Continue"
                android:background="@drawable/bg_primary_button"
                android:textColor="#FFFFFF"/>

        </LinearLayout>

        <TextView
            android:id="@+id/textWelcome"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text=""
            android:textColor="#FFFFFF"
            android:textSize="16sp"
            android:gravity="center"
            android:layout_marginTop="18dp" />

    </LinearLayout>

</ScrollView>
```

activity_dashboard.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/fitness_background"
    android:fillViewport="true">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Smart Health &amp; Fitness"
            android:textColor="#FFFFFF"
            android:textSize="28sp"
            android:textStyle="bold"
            android:gravity="center"
            android:layout_marginTop="20dp" />

        <TextView
            android:id="@+id/textDashboardGreeting"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Welcome!"
            android:textColor="#E6FFFB"
            android:textSize="17sp"
            android:gravity="center"
            android:layout_marginTop="8dp"
            android:layout_marginBottom="24dp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/bg_white_card"
            android:layout_marginBottom="16dp">

            <TextView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="TODAY'S GOAL"
                android:textColor="#0F766E"
                android:textSize="14sp"
                android:textStyle="bold"
                android:gravity="center" />

            <TextView
                android:id="@+id/textDashboardGoal"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Goal: 0 steps"
                android:textColor="#1F2937"
                android:textSize="22sp"
                android:textStyle="bold"
                android:gravity="center"
                android:layout_marginTop="8dp" />

            <Button
                android:id="@+id/btnEditGoal"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Change Daily Goal"
                android:background="@drawable/bg_primary_button"
                android:textColor="#FFFFFF"
                android:layout_marginTop="14dp" />

        </LinearLayout>

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:orientation="vertical"
            android:background="@drawable/bg_white_card"
            android:layout_marginBottom="16dp">

            <TextView
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="YOUR MOVEMENT"
                android:textColor="#0F766E"
                android:textSize="14sp"
                android:textStyle="bold"
                android:gravity="center" />

            <TextView
                android:id="@+id/textStepCount"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="0 Steps"
                android:textColor="#1F2937"
                android:textSize="36sp"
                android:textStyle="bold"
                android:gravity="center"
                android:layout_marginTop="10dp" />

            <TextView
                android:id="@+id/textDuration"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Activity time: 0 minutes"
                android:textColor="#4B5563"
                android:textSize="16sp"
                android:gravity="center"
                android:layout_marginTop="6dp" />

            <TextView
                android:id="@+id/textSensorStatus"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Ready when you are."
                android:textColor="#4B5563"
                android:textSize="15sp"
                android:gravity="center"
                android:layout_marginTop="10dp" />

            <Button
                android:id="@+id/btnStartMonitoring"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Start Activity Monitoring"
                android:background="@drawable/bg_primary_button"
                android:textColor="#FFFFFF"
                android:layout_marginTop="14dp" />

        </LinearLayout>

        <Button
            android:id="@+id/btnSaveActivity"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Save Today's Activity"
            android:background="@drawable/bg_primary_button"
            android:textColor="#FFFFFF"
            android:layout_marginBottom="10dp" />

        <Button
            android:id="@+id/btnViewRecords"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="View Previous Records"
            android:background="@drawable/bg_primary_button"
            android:textColor="#FFFFFF"
            android:layout_marginBottom="20dp" />

    </LinearLayout>

</ScrollView>
```

activity_records.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/bg_dashboard">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">

        <TextView
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Previous Activity Records"
            android:textColor="#FFFFFF"
            android:textSize="27sp"
            android:textStyle="bold"
            android:gravity="center"
            android:layout_marginTop="24dp"
            android:layout_marginBottom="24dp" />

        <LinearLayout
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:orientation="vertical"
            android:background="@drawable/bg_white_card">

            <TextView
                android:id="@+id/textRecords"
                android:layout_width="match_parent"
                android:layout_height="wrap_content"
                android:text="Loading your activity records..."
                android:textColor="#1F2937"
                android:textSize="17sp"
                android:lineSpacingExtra="10dp" />

        </LinearLayout>

        <Button
            android:id="@+id/btnBackDashboard"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Back to Dashboard"
            android:background="@drawable/bg_primary_button"
            android:textColor="#FFFFFF"
            android:layout_marginTop="20dp" />

    </LinearLayout>

</ScrollView>
```

bg_primary_button.xml

```
<?xml version="1.0" encoding="utf-8"?>
<selector xmlns:android="http://schemas.android.com/apk/res/android">

    <item>
        <shape android:shape="rectangle">
            <solid android:color="#0F766E" />
            <corners android:radius="28dp" />
            <padding
                android:left="16dp"
                android:top="10dp"
                android:right="16dp"
                android:bottom="10dp" />
        </shape>
    </item>

</selector>
```

bg_white_card.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">

    <solid android:color="#F9FFFE" />

    <corners android:radius="24dp" />

    <padding
        android:left="20dp"
        android:top="20dp"
        android:right="20dp"
        android:bottom="20dp" />

</shape>
```

bg_dashboard.xml

```
<?xml version="1.0" encoding="utf-8"?>
<shape xmlns:android="http://schemas.android.com/apk/res/android"
    android:shape="rectangle">

    <gradient
        android:angle="135"
        android:startColor="#0F766E"
        android:centerColor="#14B8A6"
        android:endColor="#99F6E4" />

</shape>
```

colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

strings.xml

```
<resources>
    <string name="app_name">SmartHealthFitnessMonitor</string>
</resources>
```

themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Base.Theme.SmartHealthFitnessMonitor" parent="Theme.Material3.DayNight.NoActionBar">
        <!-- Customize your light theme here. -->
        <!-- <item name="colorPrimary">@color/my_light_primary</item> -->
    </style>

    <style name="Theme.SmartHealthFitnessMonitor" parent="Base.Theme.SmartHealthFitnessMonitor" />
</resources>
```

7. Conclusion and Future Enhancements

The Smart Health and Fitness Monitor app successfully implements the required mini-project functionality. It manages user information, estimates activity from accelerometer data, stores activity records locally in SQLite, preserves a user-selected daily goal, and provides a motivational notification when the goal is achieved. The supplied screenshots show these features working on an Android device.

Possible Future Enhancements

- Use the device step-counter sensor when available to improve step-count accuracy.
- Add a calendar and charts to show weekly or monthly activity trends.
- Allow a user to edit or delete an incorrect activity record.
- Add optional reminders when there has been little movement during the day.
- Support secure cloud backup or account-based synchronization in a future version.